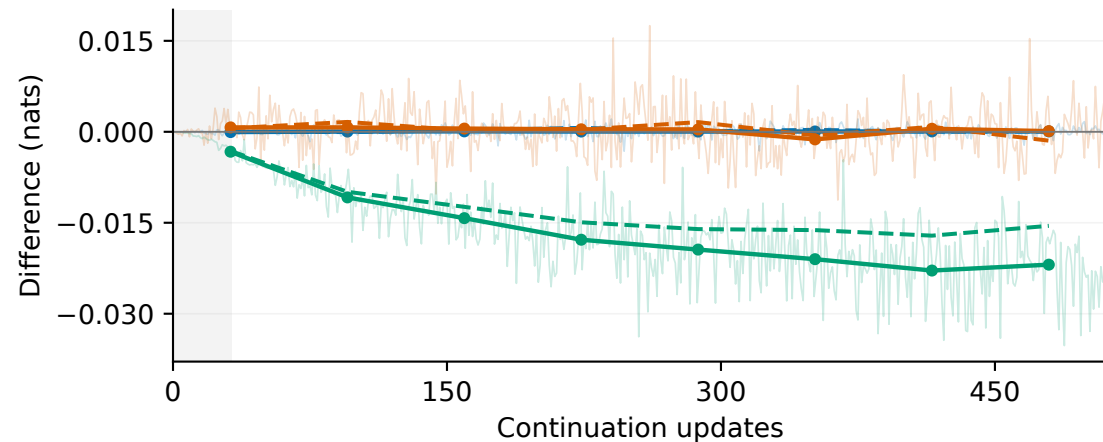
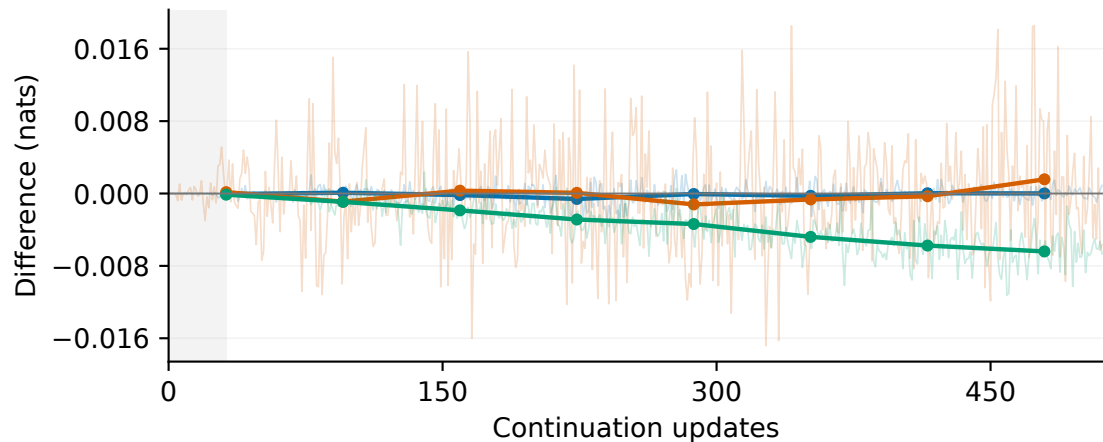


—●— SGD – Heavy Ball effective $h=3e-06$
—●— SGD – Heavy Ball effective $h=3e-05$
—●— Adam – Head only (different settings) raw $\eta=2.9e-07 / 3e-06$

(a) Paired measured response and reconstruction



(b) Paired remaining response $\Delta\epsilon$



Light lines: observed values. Dots: 64-step means. Panel (a): solid ΔL ; dashed $\Delta A + \Delta Q_2$.
 Common difficulty cancels. Original shared step axis; no fitted time alignment.
 Rate-refinement protocol (adaptive).